

The dyadic Goldbach rate and the Riemann hypothesis are one proposition

A self-contained proof of the biconditional

Abstract

Let Λ be the von Mangoldt function, $R(N) = \sum_{a+b=N} \Lambda(a)\Lambda(b)$ the quantitative Goldbach count, and $\mathcal{G}(t) = t^2 \sum_N R(N)e^{-Nt}$ its normalised Laplace transform. Write **G** for the proposition “for every $\varepsilon > 0$, $\mathcal{G}(2t) - \mathcal{G}(t) = O_\varepsilon(t^{1/2-\varepsilon})$ as $t \downarrow 0$ ” and RH for the Riemann hypothesis. We prove $\mathbf{G} \Leftrightarrow \text{RH}$. Hence any proof of either proposition is a proof of the other, and any refutation of either is a refutation of the other: the two have identical truth content. Every fact about ζ , Γ and the primes that the argument uses is proved in this file. The only inputs are the standard theorems of real and complex analysis listed in Section 2.

1 Statement

Definition 1.1 (Arithmetic objects). For $n \geq 1$ let $\Lambda(n) = \log p$ if $n = p^k$ for a prime p and an integer $k \geq 1$, and $\Lambda(n) = 0$ otherwise. Let $\psi(x) = \sum_{n \leq x} \Lambda(n)$ for real $x \geq 1$. For $N \geq 2$ let

$$R(N) = \sum_{\substack{a+b=N \\ a,b \geq 1}} \Lambda(a)\Lambda(b),$$

the weighted ordered count of representations of N as a sum of two prime powers. For $t > 0$ let

$$A(t) = \sum_{n \geq 1} \Lambda(n)e^{-nt}, \quad G_R(t) = \sum_{N \geq 2} R(N)e^{-Nt}, \quad \mathcal{G}(t) = t^2 G_R(t), \quad r(t) = \mathcal{G}(2t) - \mathcal{G}(t).$$

All series converge absolutely for $t > 0$ since $0 \leq \Lambda(n) \leq \log n \leq n$ and $R(N) \leq N^2 \cdot N = N^3$.

Lemma 1.2. *For every $t > 0$, $G_R(t) = A(t)^2$; hence $\mathcal{G}(t) = (tA(t))^2$.*

Proof. All terms are nonnegative, so by Tonelli

$$A(t)^2 = \sum_{a \geq 1} \sum_{b \geq 1} \Lambda(a)\Lambda(b)e^{-(a+b)t} = \sum_{N \geq 2} e^{-Nt} \sum_{a+b=N} \Lambda(a)\Lambda(b). \quad \square$$

Definition 1.3 (The two propositions). **G**: For every $\varepsilon \in (0, \frac{1}{2})$ there exist $C_\varepsilon > 0$ and $t_\varepsilon > 0$ such that $|r(t)| \leq C_\varepsilon t^{1/2-\varepsilon}$ for all $0 < t \leq t_\varepsilon$.

RH: The Riemann zeta function $\zeta(s) = \sum_{n \geq 1} n^{-s}$ ($\text{Re } s > 1$) has a meromorphic continuation to $\mathbb{C}$ (Theorem 4.5 below), and every zero ρ of this continuation with $0 < \text{Re } \rho < 1$ satisfies $\text{Re } \rho = \frac{1}{2}$.

Theorem 1.4 (Main theorem). $\mathbf{G} \Leftrightarrow \text{RH}$.

Corollary 1.5. $\neg G \iff \neg RH$. A proof of either proposition yields, through Sections 6 and 7, a proof of the other; a counterexample to either yields a refutation of the other.

Proof. Immediate from Theorem 1.4, constructively: from $P \leftrightarrow Q$ one obtains $\neg P \leftrightarrow \neg Q$ by composing with the implications. $\square$

Remark 1.6. G is a statement about the quantitative Goldbach field $R(N)$ through its Laplace transform. Section 7 shows that the finite prefix $R(N)$, $N \leq L$, determines the quantity at scale t up to an error smaller than the critical rate once $L \gg t^{-1} \log(1/t)$, so G is a statement about finite Goldbach data at every scale. Nothing in this file concerns the proposition “ $R(N) > 0$ for all even $N \geq 4$ ”, which is a different proposition.

2 Inputs from analysis

The proof uses only the following standard results; nothing else is assumed.

- (A1) Cauchy’s theorem and the residue theorem for rectangles; Cauchy’s estimate $|g'(z)| \leq \max_{|w-z|=\delta} |g(w)|/\delta$.
- (A2) The identity theorem: two meromorphic functions on a connected open set that agree on a nonempty open subset agree everywhere. A nonvanishing holomorphic function on a convex open set has a holomorphic logarithm. The logarithmic derivative of a meromorphic function h has simple poles exactly at the zeros and poles of h , with residue the order (positive at zeros, negative at poles).
- (A3) Holomorphic dependence on parameters: if $f(s, x)$ is holomorphic in s for each x , measurable in x , and $|f(s, x)| \leq g(x)$ with $\int g < \infty$ for s in a neighbourhood of each point, then $s \mapsto \int f(s, x) dx$ is holomorphic there. A locally uniformly convergent series of holomorphic functions is holomorphic and may be differentiated termwise.
- (A4) Fubini and Tonelli.
- (A5) For a Schwartz function φ on $\mathbb{R}$ with $\widehat{\varphi}(\xi) = \int_{\mathbb{R}} \varphi(u) e^{-2\pi i u \xi} du$: Fourier inversion $\varphi(u) = \int_{\mathbb{R}} \widehat{\varphi}(\xi) e^{2\pi i u \xi} d\xi$, and Poisson summation $\sum_{n \in \mathbb{Z}} \varphi(n) = \sum_{n \in \mathbb{Z}} \widehat{\varphi}(n)$.
- (A6) The Schwarz lemma and the maximum principle.
- (A7) Unique factorisation of positive integers into primes.

3 The Gamma function

Definition 3.1. For $\operatorname{Re} s > 0$ let $\Gamma(s) = \int_0^\infty e^{-x} x^{s-1} dx$.

Lemma 3.2. Γ is holomorphic on $\operatorname{Re} s > 0$, $\Gamma(s+1) = s\Gamma(s)$, $\Gamma(1) = 1$, $\Gamma(n) = (n-1)!$ for $n \geq 1$. Γ extends to a meromorphic function on $\mathbb{C}$ whose poles are exactly the simple poles at $0, -1, -2, \dots$

Proof. Holomorphy by (A3) with $g(x) = e^{-x}(x^{a-1} + x^{b-1})$ on $a \leq \operatorname{Re} s \leq b$. Integration by parts gives $\Gamma(s+1) = s\Gamma(s)$, and $\Gamma(1) = 1$. For $\operatorname{Re} s > -n$ define $\Gamma(s) = \Gamma(s+n)/(s(s+1)\cdots(s+n-1))$; the definitions agree on overlaps by the functional equation. The only singularities are at $s = -j$, $j \geq 0$: choosing $n > j$, the numerator there is $\Gamma(n-j) = (n-j-1)! \neq 0$ while the denominator has a simple zero, so each is a simple pole. $\square$

Lemma 3.3 (Beta integral). *For $\operatorname{Re} a > 0$, $\operatorname{Re} b > 0$: $\Gamma(a)\Gamma(b) = \Gamma(a+b) \int_0^1 v^{a-1}(1-v)^{b-1} dv$.*

Proof. $\Gamma(a)\Gamma(b) = \iint_{(0,\infty)^2} e^{-x-y} x^{a-1} y^{b-1} dx dy$ (absolutely convergent, (A4)). Substitute $x = wv$, $y = w(1-v)$ with $w > 0$, $0 < v < 1$; the Jacobian is w , and the integral becomes $\int_0^\infty e^{-w} w^{a+b-1} dw \int_0^1 v^{a-1}(1-v)^{b-1} dv$. $\square$

Lemma 3.4 (Reflection). *For all $s \in \mathbb{C} \setminus \mathbb{Z}$: $\Gamma(s)\Gamma(1-s) = \pi/\sin(\pi s)$.*

Proof. First let $0 < \operatorname{Re} s < 1$. By Lemma 3.3 with $a = s$, $b = 1-s$, $\Gamma(s)\Gamma(1-s) = \int_0^1 v^{s-1}(1-v)^{-s} dv$. Substituting $v = u/(1+u)$, $dv = du/(1+u)^2$, $v^{s-1}(1-v)^{-s} = u^{s-1}(1+u)^{-s}$, this equals $I(s) = \int_0^\infty \frac{u^{s-1}}{1+u} du = \int_{\mathbb{R}} \frac{e^{sv}}{1+e^v} dv$ (with $u = e^v$). Let $g(z) = e^{sz}/(1+e^z)$ and integrate over the rectangle with vertices $\pm T$, $\pm T+2\pi i$. Inside, g has exactly one pole, at $z = i\pi$, where $1+e^z$ vanishes with derivative $e^{i\pi} = -1$, so $\operatorname{Res}_{i\pi} g = -e^{i\pi s}$. The top edge contributes $-\int_{-T}^T g(v+2\pi i) dv = -e^{2\pi i s} \int_{-T}^T g(v) dv$. On the right edge $|g(T+iy)| \leq e^{\sigma T}/(e^T-1) \rightarrow 0$ since $\sigma = \operatorname{Re} s < 1$; on the left edge $|g(-T+iy)| \leq e^{-\sigma T}/(1-e^{-T}) \rightarrow 0$ since $\sigma > 0$. Letting $T \rightarrow \infty$: $(1-e^{2\pi i s})I(s) = 2\pi i(-e^{i\pi s})$, so $I(s) = \frac{2\pi i}{e^{i\pi s} - e^{-i\pi s}} = \frac{\pi}{\sin \pi s}$. Both sides of the claimed identity are meromorphic on $\mathbb{C}$ and agree on the strip, so they agree on $\mathbb{C} \setminus \mathbb{Z}$ by (A2). $\square$

Corollary 3.5. $\Gamma(s) \neq 0$ for every $s \in \mathbb{C}$ that is not a pole; in particular $\Gamma(s) \neq 0$ for $\operatorname{Re} s > 0$, and $1/\Gamma$ is entire.

Proof. If $s \notin \mathbb{Z}$ then $\Gamma(s)\Gamma(1-s) = \pi/\sin \pi s$ is finite and nonzero, so $\Gamma(s) \neq 0$. If $s = n \geq 1$ then $\Gamma(n) = (n-1)! \neq 0$. Nonpositive integers are poles. Hence $1/\Gamma$ has no poles, and its zeros are the poles of Γ . $\square$

Lemma 3.6 (Decay on vertical strips). *For $0 < a \leq b < \infty$ there is $K = K(a, b)$ with $|\Gamma(\sigma+i\tau)| \leq K/(1+\tau^2)$ for all $a \leq \sigma \leq b$, $\tau \in \mathbb{R}$.*

Proof. Substituting $x = e^u$, $\Gamma(\sigma+i\tau) = \int_{\mathbb{R}} \varphi_\sigma(u) e^{i\tau u} du$ with $\varphi_\sigma(u) = \exp(\sigma u - e^u)$, a Schwartz function. Two integrations by parts give, for $\tau \neq 0$, $\Gamma(\sigma+i\tau) = -\tau^{-2} \int_{\mathbb{R}} \varphi''_\sigma(u) e^{i\tau u} du$, and $\varphi''_\sigma = ((\sigma - e^u)^2 - e^u) \varphi_\sigma$, so

$$|\Gamma(\sigma+i\tau)| \leq \tau^{-2} \int_{\mathbb{R}} ((\sigma - e^u)^2 + e^u) e^{\sigma u - e^u} du = \tau^{-2} (\Gamma(\sigma+2) - (2\sigma-1)\Gamma(\sigma+1) + \sigma^2\Gamma(\sigma)),$$

using $\int_{\mathbb{R}} e^{ku} e^{\sigma u - e^u} du = \Gamma(\sigma+k)$. The right side is a continuous function of $\sigma \in [a, b]$, bounded by some K_1 ; also $|\Gamma(\sigma+i\tau)| \leq \Gamma(\sigma) \leq K_2$ on $[a, b]$. Take $K = 2(K_1 + K_2)$. $\square$

Lemma 3.7 (Cahen–Mellin integral). *For $c > 0$ and $x > 0$: $e^{-x} = \frac{1}{2\pi} \int_{\mathbb{R}} \Gamma(c+i\tau) x^{-c-i\tau} d\tau$.*

Proof. With $\varphi_c(u) = \exp(cu - e^u)$ as above, $\Gamma(c+i\tau) = \int_{\mathbb{R}} \varphi_c(u) e^{i\tau u} du = \widehat{\varphi_c}(-\tau/2\pi)$. Fourier inversion (A5) gives $\varphi_c(u) = \frac{1}{2\pi} \int_{\mathbb{R}} \Gamma(c+i\tau) e^{-i\tau u} d\tau$. Put $u = \log x$: $x^c e^{-x} = \frac{1}{2\pi} \int_{\mathbb{R}} \Gamma(c+i\tau) x^{-c-i\tau} d\tau$. $\square$

4 The zeta function

Lemma 4.1 (Euler product and logarithm). *For $\operatorname{Re} s > 1$ the series $L(s) = \sum_p \sum_{k \geq 1} \frac{p^{-ks}}{k} = \sum_{n \geq 2} \frac{\Lambda(n)}{\log n} n^{-s}$ converges absolutely and locally uniformly, $\zeta(s) = \exp L(s)$, hence $\zeta(s) \neq 0$, and*

$$-\frac{\zeta'}{\zeta}(s) = \sum_{n \geq 1} \Lambda(n) n^{-s} \quad (\operatorname{Re} s > 1).$$

Moreover $|L(2 + i\tau)| \leq L(2) = \log \zeta(2) < 1$ for all real τ .

Proof. The two expressions for L agree termwise: for $n = p^k$, $1/k = \log p / \log n = \Lambda(n) / \log n$. Absolute convergence: $\sum_p \sum_k |p^{-ks}|/k \leq \sum_{n \geq 2} n^{-\sigma} < \infty$ for $\sigma = \operatorname{Re} s > 1$, locally uniformly, so L is holomorphic (A3). For a fixed prime p , $\exp(\sum_k p^{-ks}/k) = (1 - p^{-s})^{-1}$ by the power series of $-\log(1 - z)$, $|z| < 1$. Hence for a bound P , $\exp(\sum_{p \leq P} \sum_k p^{-ks}/k) = \prod_{p \leq P} \sum_{k \geq 0} p^{-ks} = \sum_{n \in S_P} n^{-s}$, where S_P is the set of positive integers all of whose prime factors are at most P ; the last equality is (A7), the expansion of the finite product of absolutely convergent series. As $P \rightarrow \infty$ the left side tends to $\exp L(s)$ and the right side to $\zeta(s)$ (the omitted terms are bounded by $\sum_{n > P} n^{-\sigma} \rightarrow 0$). Termwise differentiation (A3): $L'(s) = -\sum_p \sum_k (\log p) p^{-ks} = -\sum_n \Lambda(n) n^{-s}$, and $\zeta'/\zeta = L'$. Finally $|L(2 + i\tau)| \leq \sum_p \sum_k p^{-2k}/k = L(2) = \log \zeta(2)$ and $\zeta(2) \leq 1 + \int_1^\infty x^{-2} dx = 2 < e$. $\square$

Lemma 4.2 (Continuation to $\operatorname{Re} s > 0$ and a growth bound). *For every integer $N \geq 1$ and every s with $\operatorname{Re} s > 0$, $s \neq 1$,*

$$\zeta(s) = \sum_{n \leq N} n^{-s} + \frac{N^{1-s}}{s-1} - s \int_N^\infty \{x\} x^{-s-1} dx, \quad (1)$$

where $\{x\} = x - \lfloor x \rfloor$; the right side defines a meromorphic continuation of ζ to $\operatorname{Re} s > 0$ whose only pole is simple, at $s = 1$, with residue 1. Consequently $-\zeta'/\zeta$ is meromorphic on $\operatorname{Re} s > 0$, with a simple pole of residue 1 at $s = 1$ and simple poles of residue $-m$ at each zero of ζ of multiplicity m , and no other poles. Moreover

$$|\zeta(\sigma + i\tau)| \leq 12\sqrt{|\tau|} \quad \text{for } \frac{1}{2} \leq \sigma \leq 4, |\tau| \geq 2. \quad (2)$$

Proof. For $\operatorname{Re} s > 1$, Abel summation gives $\sum_{n > N} n^{-s} = \int_N^\infty x^{-s} d[x] = -N^{1-s} + s \int_N^\infty [x] x^{-s-1} dx$, and writing $[x] = x - \{x\}$, $s \int_N^\infty x^{-s} dx = sN^{1-s}/(s-1)$, so $\sum_{n > N} n^{-s} = N^{1-s}/(s-1) - s \int_N^\infty \{x\} x^{-s-1} dx$, which is (1). The integral converges absolutely for $\operatorname{Re} s > 0$, dominated locally uniformly by $\int_N^\infty x^{-\sigma_0-1} dx$, hence is holomorphic there (A3); the continuation follows by (A2). With $N = 1$, $\zeta(s) = \frac{s}{s-1} - s \int_1^\infty \{x\} x^{-s-1} dx$, so $(s-1)\zeta(s) \rightarrow 1$ as $s \rightarrow 1$. The statement about $-\zeta'/\zeta$ is (A2).

For (2): take $N = \lfloor |\tau| \rfloor$, so $|\tau|/2 \leq N \leq |\tau|$. Then $|\sum_{n \leq N} n^{-s}| \leq \sum_{n \leq N} n^{-1/2} \leq 2\sqrt{N} \leq 2\sqrt{|\tau|}$; $|N^{1-s}/(s-1)| \leq N^{1/2}/|\tau| \leq 1$; and $|s \int_N^\infty \{x\} x^{-s-1} dx| \leq |s| N^{-\sigma}/\sigma \leq (4 + |\tau|) \cdot 2 \cdot \sqrt{2/|\tau|} \leq 9\sqrt{|\tau|}$ for $|\tau| \geq 2$. $\square$

Lemma 4.3 (No zeros on $\operatorname{Re} s = 1$). $\zeta(1 + i\tau) \neq 0$ for every real $\tau \neq 0$. Together with Lemma 4.1, $\zeta(s) \neq 0$ for $\operatorname{Re} s \geq 1$, $s \neq 1$.

Proof. For $\sigma > 1$ and real τ , by Lemma 4.1, $\operatorname{Re} L(\sigma + i\tau) = \sum_{n \geq 2} \frac{\Lambda(n)}{\log n} n^{-\sigma} \cos(\tau \log n)$. Since $3 + 4 \cos \theta + \cos 2\theta = 2(1 + \cos \theta)^2 \geq 0$,

$$3L(\sigma) + 4 \operatorname{Re} L(\sigma + i\tau) + \operatorname{Re} L(\sigma + 2i\tau) \geq 0, \quad \text{so} \quad \zeta(\sigma)^3 |\zeta(\sigma + i\tau)|^4 |\zeta(\sigma + 2i\tau)| \geq 1.$$

Suppose $\zeta(1+i\tau_0) = 0$ with $\tau_0 \neq 0$. By Lemma 4.2, ζ is holomorphic near $1+i\tau_0$ and near $1+2i\tau_0$, so there are constants with $|\zeta(\sigma+i\tau_0)| \leq K(\sigma-1)$ and $|\zeta(\sigma+2i\tau_0)| \leq K'$ for $1 < \sigma \leq 2$, while $\zeta(\sigma) = \sigma/(\sigma-1) - \sigma \int_1^\infty \{x\}x^{-\sigma-1} dx \leq \sigma/(\sigma-1) \leq 2/(\sigma-1)$ by (1) with $N=1$, the subtracted integral being nonnegative for real $\sigma > 1$. Then the product is at most $8K^4K'(\sigma-1) \rightarrow 0$ as $\sigma \downarrow 1$, a contradiction. $\square$

Lemma 4.4 (Gaussian transform and theta inversion). *For $x > 0$ let $\theta(x) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 x}$ and $\omega(x) = \sum_{n \geq 1} e^{-\pi n^2 x}$, so $\theta = 1 + 2\omega$. Then $\theta(1/x) = \sqrt{x}\theta(x)$, and $0 < \omega(x) \leq 2e^{-\pi x}$ for $x \geq 1$.*

Proof. Let $h(\xi) = \int_{\mathbb{R}} e^{-\pi u^2} e^{-2\pi i u \xi} du$. Differentiating under the integral and integrating by parts, $h'(\xi) = \int_{\mathbb{R}} (-2\pi i u) e^{-\pi u^2} e^{-2\pi i u \xi} du = i \int_{\mathbb{R}} (e^{-\pi u^2})' e^{-2\pi i u \xi} du = -2\pi \xi h(\xi)$, so $h(\xi) = h(0)e^{-\pi \xi^2} = e^{-\pi \xi^2}$. Scaling $u \mapsto u\sqrt{x}$, the function $f(u) = e^{-\pi x u^2}$ has $\widehat{f}(\xi) = x^{-1/2} e^{-\pi \xi^2/x}$. Poisson summation (A5) gives $\theta(x) = x^{-1/2} \theta(1/x)$. For $x \geq 1$: $\omega(x) \leq \sum_{n \geq 1} e^{-\pi n^2 x} = e^{-\pi x}/(1 - e^{-\pi x}) \leq 2e^{-\pi x}$. $\square$

Theorem 4.5 (Functional equation). *Let $\xi(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$ for $\operatorname{Re} s > 1$. Then*

$$\xi(s) = \frac{1}{s(s-1)} + \int_1^\infty \omega(x) (x^{s/2-1} + x^{(1-s)/2-1}) dx, \quad (3)$$

where the integral is an entire function of s . Consequently ζ extends to a meromorphic function on $\mathbb{C}$, agreeing with Lemma 4.2 on $\operatorname{Re} s > 0$, and

$$\pi^{-s/2} \Gamma(s/2) \zeta(s) = \pi^{-(1-s)/2} \Gamma((1-s)/2) \zeta(1-s)$$

as meromorphic functions on $\mathbb{C}$.

Proof. For $\operatorname{Re} s > 1$ and $n \geq 1$, substituting $y = \pi n^2 x$ gives $\int_0^\infty e^{-\pi n^2 x} x^{s/2-1} dx = (\pi n^2)^{-s/2} \Gamma(s/2)$. Summing over n and interchanging, which is legitimate because

$$\sum_{n \geq 1} \int_0^\infty e^{-\pi n^2 x} x^{\sigma/2-1} dx = \pi^{-\sigma/2} \Gamma(\sigma/2) \zeta(\sigma) < \infty, \quad \text{gives} \quad \xi(s) = \int_0^\infty \omega(x) x^{s/2-1} dx.$$

Split at $x = 1$. In $\int_0^1$ substitute $x = 1/y$ and use $\omega(1/y) = -\frac{1}{2} + \frac{\sqrt{y}}{2} + \sqrt{y}\omega(y)$ (from Lemma 4.4):

$$\int_0^1 \omega(x) x^{s/2-1} dx = \int_1^\infty \omega(1/y) y^{-s/2-1} dy = -\frac{1}{s} + \frac{1}{s-1} + \int_1^\infty \omega(y) y^{(1-s)/2-1} dy,$$

using $\int_1^\infty y^{-s/2-1} dy = 2/s$ and $\int_1^\infty y^{-s/2-1/2} dy = 2/(s-1)$ for $\operatorname{Re} s > 1$. Since $-1/s + 1/(s-1) = 1/(s(s-1))$, (3) follows. The integral in (3) is entire by (A3), since $\omega(x) \leq 2e^{-\pi x}$ dominates $x^{\sigma/2-1} + x^{(1-\sigma)/2-1}$ times $e^{-\pi x}$ locally uniformly. The right side of (3) is invariant under $s \mapsto 1-s$, which is the functional equation for ξ . Define $\zeta(s) = \pi^{s/2} \xi(s)/\Gamma(s/2)$; by Corollary 3.5 the factor $1/\Gamma(s/2)$ is entire, so ζ is meromorphic on $\mathbb{C}$, and it agrees with the Dirichlet series on $\operatorname{Re} s > 1$, hence with Lemma 4.2 on $\operatorname{Re} s > 0$ by (A2). $\square$

Corollary 4.6 (Symmetry of the zeros). *For $0 < \operatorname{Re} s < 1$: $\zeta(s) = 0 \iff \zeta(1-s) = 0$.*

Proof. If $0 < \operatorname{Re} s < 1$ then $\operatorname{Re}(s/2) > 0$ and $\operatorname{Re}((1-s)/2) > 0$, so $\Gamma(s/2)$ and $\Gamma((1-s)/2)$ are finite and nonzero (Corollary 3.5), as are the powers of π . Apply Theorem 4.5. $\square$

5 A lower bound for ψ

Lemma 5.1. *For every integer $n \geq 1$, $\psi(n) = \log \text{lcm}(1, 2, \dots, n)$.*

Proof. By (A7), $\text{lcm}(1, \dots, n) = \prod_{p \leq n} p^{k_p}$ where $k_p = \max\{k : p^k \leq n\}$, and $\log$ of this is $\sum_{p \leq n} k_p \log p = \sum_{p^k \leq n} \log p = \psi(n)$. $\square$

Lemma 5.2. *For integers $1 \leq m \leq n$, $m \binom{n}{m}$ divides $\text{lcm}(1, \dots, n)$. Consequently $\psi(2m) \geq 2m \log 2 - \log 3$ for $m \geq 1$, and $\psi(x) \geq x \log 2 - 3$ for all real $x \geq 2$.*

Proof. Let $I = \int_0^1 v^{m-1} (1-v)^{n-m} dv$. Expanding $(1-v)^{n-m}$ by the binomial theorem, $I = \sum_{j=0}^{n-m} \binom{n-m}{j} \frac{(-1)^j}{m+j}$, so $\text{lcm}(1, \dots, n) \cdot I$ is an integer; it is positive since the integrand is positive. By Lemma 3.3 with $a = m$, $b = n - m + 1$ and Lemma 3.2, $I = \frac{(m-1)!(n-m)!}{n!} = \frac{1}{m \binom{n}{m}}$. Hence $\text{lcm}(1, \dots, n) \cdot I = \text{lcm}(1, \dots, n) / (m \binom{n}{m})$ is a positive integer, which is the divisibility claim. Take $n = 2m$: $\text{lcm}(1, \dots, 2m) \geq m \binom{2m}{m} \geq m \cdot \frac{4^m}{2m+1} \geq \frac{4^m}{3}$, because $4^m = \sum_{j=0}^{2m} \binom{2m}{j} \leq (2m+1) \binom{2m}{m}$ and $m/(2m+1) \geq 1/3$. By Lemma 5.1, $\psi(2m) \geq 2m \log 2 - \log 3$. For real $x \geq 2$ let $m = \lfloor x/2 \rfloor \geq 1$; then $\psi(x) \geq \psi(2m) \geq (x-2) \log 2 - \log 3 \geq x \log 2 - 3$. $\square$

Lemma 5.3. *Let $Y(t) = tA(t)$ and $c_0 = \frac{\log 2}{2e}$, $t_0 = \frac{\log 2}{6}$. Then $Y(t) \geq c_0$ for all $0 < t \leq t_0$.*

Proof. For $0 < t \leq \frac{1}{2}$, $A(t) \geq \sum_{n \leq 1/t} \Lambda(n) e^{-nt} \geq e^{-1} \psi(1/t) \geq e^{-1} (\frac{\log 2}{t} - 3)$ by Lemma 5.2, so $Y(t) \geq e^{-1} (\log 2 - 3t) \geq e^{-1} \cdot \frac{\log 2}{2}$ for $t \leq t_0$. $\square$

6 Forward direction: $\text{RH} \Rightarrow \text{G}$

Throughout this section assume RH. By Lemmas 4.1 and 4.3, RH is equivalent to

$$\zeta(s) \neq 0 \quad \text{for all } s \text{ with } \text{Re } s > \frac{1}{2}. \quad (4)$$

Lemma 6.1 (Mellin representation of A). *For $t > 0$ and $c > 1$: $A(t) = \frac{1}{2\pi} \int_{\mathbb{R}} \Gamma(c + i\tau) \left(-\frac{\zeta'}{\zeta}\right)(c + i\tau) t^{-c-i\tau} d\tau$.*

Proof. By Lemma 3.7 with $x = nt$, $e^{-nt} = \frac{1}{2\pi} \int_{\mathbb{R}} \Gamma(c + i\tau) n^{-c-i\tau} t^{-c-i\tau} d\tau$. Multiply by $\Lambda(n)$ and sum. The interchange of sum and integral is justified by (A4), since $\sum_n \Lambda(n) n^{-c} \int_{\mathbb{R}} |\Gamma(c + i\tau)| d\tau < \infty$ by Lemmas 4.1 and 3.6. The inner sum is $-\zeta'/\zeta(c + i\tau)$ by Lemma 4.1. $\square$

Lemma 6.2 (Borel–Carathéodory). *Let f be holomorphic on an open set containing the closed disc $|z| \leq R$, with $\text{Re } f(z) \leq M$ for $|z| \leq R$. Then for $0 < r < R$ and $|z| \leq r$,*

$$|f'(z)| \leq \frac{4(R+r)}{(R-r)^2} (M - \text{Re } f(0)).$$

Proof. Let $g = f - f(0)$ and $M_0 = M - \text{Re } f(0) \geq \text{Re } g(0) = 0$. If $M_0 = 0$ then $\text{Re } g \leq 0 = \text{Re } g(0)$, so $\text{Re } g$ attains an interior maximum and g is constant (A6); the bound is trivial. If $M_0 > 0$, put $\phi = g/(2M_0 - g)$; the denominator has real part $\geq M_0 > 0$, and $|g| \leq |2M_0 - g|$ because this is equivalent to $\text{Re } g \leq M_0$. So ϕ maps the disc into the closed unit disc with $\phi(0) = 0$, and the Schwarz lemma gives $|\phi(z)| \leq |z|/R$. Then $|g(z)| \leq \frac{|z|}{R} (2M_0 + |g(z)|)$, so $|g(z)| \leq \frac{2M_0|z|}{R-|z|}$. On $|w| \leq R_1 = \frac{R+r}{2}$ this gives $|g(w)| \leq \frac{2M_0(R+r)}{R-r}$, and Cauchy's estimate (A1) on the disc of radius $\frac{R-r}{2}$ about z ($|z| \leq r$) gives $|g'(z)| \leq \frac{2M_0(R+r)}{R-r} \cdot \frac{2}{R-r}$. $\square$

Lemma 6.3 (Conditional bound for ζ'/ζ). *Assume (4) and fix $\varepsilon \in (0, \frac{1}{2})$. Then for $\frac{1}{2} + \varepsilon \leq \sigma \leq 2$ and $|\tau| \geq 4$,*

$$\left| \frac{\zeta'}{\zeta}(\sigma + i\tau) \right| \leq \frac{144}{\varepsilon^2} \log(|\tau| + 2).$$

Proof. Fix τ with $|\tau| \geq 4$ and let $R' = \frac{3}{2} - \frac{\varepsilon}{4}$, $R = \frac{3}{2} - \frac{\varepsilon}{2}$, $r = \frac{3}{2} - \varepsilon$. Every s with $|s - (2 + i\tau)| < R'$ has $\operatorname{Re} s > \frac{1}{2} + \frac{\varepsilon}{4}$ and $|\operatorname{Im} s| > |\tau| - \frac{3}{2} \geq \frac{5}{2}$, so by (4) and Lemma 4.2 ζ is holomorphic and nonvanishing on this open disc. By (A2) there is a holomorphic f on $|z| < R'$ with $e^{f(z)} = \zeta(2 + i\tau + z)$; adjusting by a constant $2\pi ik$ we may take $f(0) = L(2 + i\tau)$ (Lemma 4.1), so $|f(0)| \leq 1$. For $|z| \leq R$ the point $s = 2 + i\tau + z$ has $\frac{1}{2} \leq \operatorname{Re} s \leq 4$ and $2 \leq |\operatorname{Im} s| \leq |\tau| + 2$, so by (2) $\operatorname{Re} f(z) = \log |\zeta(s)| \leq \log 12 + \frac{1}{2} \log(|\tau| + 2) =: M$. Lemma 6.2 gives, for $|z| \leq r$,

$$|f'(z)| \leq \frac{4(R+r)}{(R-r)^2} (M - \operatorname{Re} f(0)) \leq \frac{4(R+r)}{(R-r)^2} (M+1).$$

Here $R - r = \varepsilon/2$ and $R + r \leq 3$, so $4(R+r)/(R-r)^2 \leq 48/\varepsilon^2$. Since $|\tau| \geq 4$, $\log(|\tau| + 2) \geq \log 6 > 1.79$, hence $M + 1 = \log 12 + 1 + \frac{1}{2} \log(|\tau| + 2) \leq (\frac{\log 12 + 1}{1.79} + \frac{1}{2}) \log(|\tau| + 2) \leq 3 \log(|\tau| + 2)$. Therefore $|f'(z)| \leq \frac{144}{\varepsilon^2} \log(|\tau| + 2)$ for $|z| \leq r$. For $\frac{1}{2} + \varepsilon \leq \sigma \leq 2$ the point $z = \sigma - 2$ satisfies $|z| \leq \frac{3}{2} - \varepsilon = r$, and $f'(z) = \zeta'/\zeta(\sigma + i\tau)$. $\square$

Theorem 6.4. *Assume RH. For every $\varepsilon \in (0, \frac{1}{2})$ there is K_ε such that $|A(t) - 1/t| \leq K_\varepsilon t^{-1/2-\varepsilon}$ for all $t > 0$. Consequently **G** holds: $|r(t)| \leq C_\varepsilon t^{1/2-\varepsilon}$ for $0 < t \leq \frac{1}{2}$.*

Proof. Fix ε and put $c_1 = \frac{1}{2} + \varepsilon$, $c = 2$, $h(s) = \Gamma(s)(-\zeta'/\zeta)(s)t^{-s}$. On the closed strip $c_1 \leq \operatorname{Re} s \leq c$ the function h is meromorphic with a single pole, at $s = 1$ (Lemma 4.2, (4), Corollary 3.5), simple, with residue $\Gamma(1) \cdot 1 \cdot t^{-1} = 1/t$. Apply the residue theorem (A1) to the rectangle with vertices $c_1 \pm iT$, $c \pm iT$ for $T \geq 4$:

$$\int_{c-iT}^{c+iT} h \, ds - \int_{c_1-iT}^{c_1+iT} h \, ds + \int_{c_1}^c (h(\sigma - iT) - h(\sigma + iT)) \, d\sigma = \frac{2\pi i}{t}.$$

On the horizontal edges, Lemmas 3.6 and 6.3 give $|h(\sigma \pm iT)| \leq \frac{K}{1+T^2} \cdot \frac{144}{\varepsilon^2} \log(T+2) \cdot \max(t^{-c}, t^{-c_1}) \rightarrow 0$ as $T \rightarrow \infty$. By Lemma 6.1 the first integral tends to $2\pi i A(t)$. On the line $\operatorname{Re} s = c_1$, $|\zeta'/\zeta(c_1 + i\tau)|$ is bounded by $\frac{144}{\varepsilon^2} \log(|\tau| + 2)$ for $|\tau| \geq 4$ and by its maximum on the compact segment $|\tau| \leq 4$ (where it is continuous: no zero and no pole lies on $\operatorname{Re} s = c_1$), so with Lemma 3.6 $\int_{\mathbb{R}} |h(c_1 + i\tau)| \, d\tau \leq 2\pi K_\varepsilon t^{-c_1}$ for a constant K_ε independent of t . Letting $T \rightarrow \infty$: $A(t) = \frac{1}{t} + E(t)$ with $E(t) = \frac{1}{2\pi} \int_{\mathbb{R}} h(c_1 + i\tau) \, d\tau$ and $|E(t)| \leq K_\varepsilon t^{-1/2-\varepsilon}$.

For $0 < t \leq 1$ this gives $|Y(t) - 1| \leq K_\varepsilon t^{1/2-\varepsilon}$ with $Y = tA$, hence by Lemma 1.2 $|\mathcal{G}(t) - 1| = |Y(t) - 1| |Y(t) + 1| \leq K_\varepsilon (2 + K_\varepsilon) t^{1/2-\varepsilon}$, and for $0 < t \leq \frac{1}{2}$, $|r(t)| \leq |\mathcal{G}(2t) - 1| + |\mathcal{G}(t) - 1| \leq (1 + \sqrt{2})K_\varepsilon (2 + K_\varepsilon) t^{1/2-\varepsilon}$. $\square$

7 Reverse direction: **G** $\Rightarrow$ RH

Throughout this section assume **G**. Fix $\varepsilon \in (0, \frac{1}{2})$, write $\alpha = \frac{1}{2} - \varepsilon$, and let $C_\varepsilon, t_\varepsilon$ be as in **G**.

Step 1: the dyadic limit of $Y = tA$

Let c_0, t_0 be as in Lemma 5.3 and $t_1 = \min(t_0/2, t_\varepsilon)$. For $0 < t \leq t_1$ both $Y(t)$ and $Y(2t)$ are at least c_0 , so by Lemma 1.2

$$|Y(2t) - Y(t)| = \frac{|\mathcal{G}(2t) - \mathcal{G}(t)|}{Y(2t) + Y(t)} \leq \frac{C_\varepsilon}{2c_0} t^\alpha.$$

Hence for $0 < t \leq t_1$ and $K \geq 1$,

$$|Y(t) - Y(t2^{-K})| \leq \sum_{k=1}^K |Y(t2^{-k+1}) - Y(t2^{-k})| \leq \frac{C_\varepsilon}{2c_0} t^\alpha \sum_{k \geq 1} 2^{-k\alpha} = C' t^\alpha, \quad C' = \frac{C_\varepsilon}{2c_0(2^\alpha - 1)}.$$

The sequence $K \mapsto Y(t2^{-K})$ is therefore Cauchy; let $\ell(t)$ be its limit. Then for $0 < t \leq t_1$:

- (i) $|Y(t) - \ell(t)| \leq C' t^\alpha$;
- (ii) $\ell(t/2) = \ell(t)$, since both are the limit of the same sequence;
- (iii) $\ell(t) \geq c_0$, as a limit of values $\geq c_0$;
- (iv) ℓ is continuous on $(0, t_1]$: Y is continuous (the series for A converges uniformly on compact subsets of $(0, \infty)$), and $Y(t2^{-K}) \rightarrow \ell(t)$ uniformly on $(0, t_1]$ because the tail of the telescoping sum is at most $C'(t2^{-K})^\alpha \leq C' t_1^\alpha 2^{-K\alpha}$;
- (v) ℓ is bounded on $(0, t_1]$, by (ii) and continuity on $[t_1/2, t_1]$.

Consequently, for $0 < t \leq t_1$,

$$\left| A(t) - \frac{\ell(t)}{t} \right| = \frac{|Y(t) - \ell(t)|}{t} \leq C' t^{\alpha-1}. \quad (5)$$

Step 2: meromorphic continuation of $\Gamma \cdot (-\zeta'/\zeta)$ to $\operatorname{Re} s > \frac{1}{2} + \varepsilon$

For $\operatorname{Re} s > 1$, by Tonelli and $\int_0^\infty e^{-nt} t^{s-1} dt = \Gamma(s) n^{-s}$,

$$F(s) := \Gamma(s) \left(-\frac{\zeta'}{\zeta} \right)(s) = \int_0^\infty A(t) t^{s-1} dt. \quad (6)$$

Split the integral at t_1 and insert ℓ :

$$F(s) = \underbrace{\int_0^{t_1} \left(A(t) - \frac{\ell(t)}{t} \right) t^{s-1} dt}_{I_1(s)} + \underbrace{\int_0^{t_1} \ell(t) t^{s-2} dt}_{I_2(s)} + \underbrace{\int_{t_1}^\infty A(t) t^{s-1} dt}_{I_3(s)} \quad (\operatorname{Re} s > 1),$$

each piece converging absolutely for $\operatorname{Re} s > 1$ (by (5), by (v), and by $A(t) \leq \sum_n n e^{-nt} = e^{-t}(1 - e^{-t})^{-2}$).

I_1 is holomorphic on $\operatorname{Re} s > 1 - \alpha = \frac{1}{2} + \varepsilon$: by (5) the integrand is bounded by $C' t^{\alpha+\sigma-2}$, integrable on $(0, t_1]$ when $\sigma > 1 - \alpha$, locally uniformly in σ (A3).

I_3 is entire: for $t \geq t_1$, $A(t) \leq e^{-t}(1 - e^{-t_1})^{-2}$, so the integrand is dominated locally uniformly by $e^{-t} t^{\sigma-1}$ (A3).

I_2 is computed exactly from the dyadic invariance (ii). For $\operatorname{Re} s > 1$, substituting $t = 2^{-k}u$ on $[t_1 2^{-k-1}, t_1 2^{-k}]$,

$$I_2(s) = \sum_{k \geq 0} \int_{t_1 2^{-k-1}}^{t_1 2^{-k}} \ell(t) t^{s-2} dt = \sum_{k \geq 0} 2^{-k(s-1)} \Phi(s) = \frac{\Phi(s)}{1 - 2^{1-s}}, \quad \Phi(s) = \int_{t_1/2}^{t_1} \ell(u) u^{s-2} du,$$

where Φ is entire (continuous integrand on a compact interval, (A3)) and the geometric series converges since $|2^{1-s}| < 1$. The right side is meromorphic on $\mathbb{C}$, with poles contained in the set

$$Z = \{ 1 + 2\pi i k / \log 2 : k \in \mathbb{Z} \} = \{ s : 2^{1-s} = 1 \},$$

all of which lie on the line $\operatorname{Re} s = 1$.

Therefore $I_1 + \Phi/(1 - 2^{1-s}) + I_3$ is meromorphic on $\Omega_\varepsilon = \{\operatorname{Re} s > \frac{1}{2} + \varepsilon\}$ with poles in Z , and it agrees with F on $\operatorname{Re} s > 1$. Since $F = \Gamma \cdot (-\zeta'/\zeta)$ is itself meromorphic on $\operatorname{Re} s > 0$ (Lemmas 3.2 and 4.2), the identity theorem (A2) gives:

$$\text{every pole of } F \text{ in } \Omega_\varepsilon \text{ lies in } Z, \text{ hence on } \operatorname{Re} s = 1. \quad (7)$$

Step 3: no zeros of ζ in $\operatorname{Re} s > \frac{1}{2}$

Suppose $\zeta(\rho) = 0$ with $\rho \in \Omega_\varepsilon$, of multiplicity $m \geq 1$. Then $\rho \neq 1$, and by Lemma 4.2 $-\zeta'/\zeta$ has a pole at ρ with residue $-m \neq 0$. By Corollary 3.5, $\Gamma(\rho) \neq 0$ and Γ is holomorphic at ρ ($\operatorname{Re} \rho > 0$), so $F = \Gamma \cdot (-\zeta'/\zeta)$ has a pole at ρ . By (7), $\operatorname{Re} \rho = 1$. This contradicts Lemma 4.3. Hence ζ has no zero in Ω_ε . Since $\varepsilon \in (0, \frac{1}{2})$ was arbitrary, ζ has no zero with $\operatorname{Re} s > \frac{1}{2}$.

Step 4: conclusion

Let ρ be a zero of ζ with $0 < \operatorname{Re} \rho < 1$. By Step 3, $\operatorname{Re} \rho \leq \frac{1}{2}$. By Corollary 4.6, $1 - \rho$ is also a zero with $0 < \operatorname{Re}(1 - \rho) < 1$, so $\operatorname{Re}(1 - \rho) \leq \frac{1}{2}$, i.e. $\operatorname{Re} \rho \geq \frac{1}{2}$. Hence $\operatorname{Re} \rho = \frac{1}{2}$. This is RH. $\square$

Remark 7.1 (Finite data suffice at each scale). Since $0 \leq \Lambda(n) \leq \log n$, $R(N) \leq N \log^2 N$. Let $0 < t \leq e^{-3}$, $\Lambda_t = \log(1/t) \geq 3$ and $L = \lceil t^{-1} \Lambda_t \rceil$. The function $x \mapsto x \log^2 x e^{-xt}$ is decreasing for $x \geq L$ (its logarithmic derivative is $1/x + 2/(x \log x) - t < 0$ once $tx \geq 3$ and $x \geq e$), so

$$t^2 \sum_{N > L} R(N) e^{-Nt} \leq t^2 \int_L^\infty x \log^2 x e^{-xt} dx = \int_{Lt}^\infty y \log^2(y/t) e^{-y} dy \leq 2 \int_{\Lambda_t}^\infty (y^2 + y \Lambda_t^2) e^{-y} dy \leq 14 t \Lambda_t^3,$$

using $\log^2(y/t) \leq 2(\log^2 y + \Lambda_t^2)$, $\log^2 y \leq y$ for $y > 0$, and $\int_\Lambda^\infty (y^2 + y \Lambda^2) e^{-y} dy = (\Lambda^3 + 2\Lambda^2 + 2\Lambda + 2)e^{-\Lambda} \leq 7\Lambda^3 e^{-\Lambda}$ for $\Lambda \geq 1$. This is $o(t^{1/2-\varepsilon})$ for every $\varepsilon < \frac{1}{2}$. So **G** is unchanged if $\mathcal{G}(t)$ is replaced by $t^2 \sum_{N \leq L} R(N) e^{-Nt}$: at each scale t the proposition is decided, up to an error strictly below the critical rate, by the finite list $R(4), \dots, R(L)$.

8 Dependency map

Lemma 1.2	Tonelli
Lemmas 3.2–3.7	(A1), (A2), (A3), (A4), (A5), (A6): all properties of Γ used
Lemma 4.1	(A7): Euler product, $-\zeta'/\zeta = \sum \Lambda(n)n^{-s}$
Lemma 4.2	Abel summation: continuation to $\operatorname{Re} s > 0$, pole at 1, growth (2)
Lemma 4.3	3–4–1 inequality: no zeros on $\operatorname{Re} s = 1$
Lemma 4.4, Theorem 4.5, Cor. 4.6	Gaussian transform, Poisson: functional equation, zero symmetry
Lemmas 5.1–5.3	lcm and the Beta integral: $tA(t) \geq c_0$
Section 6	Mellin inversion, Borel–Carathéodory, contour shift: RH $\Rightarrow$ G
Section 7	dyadic limit, exact dyadic Mellin sum, pole bookkeeping: G $\Rightarrow$ RH

Neither direction uses the prime number theorem, zero-density estimates, the Hadamard product, or any bound on $R(N)$ beyond the trivial $R(N) \leq N \log^2 N$. The reverse direction does not need the limit ℓ to be the constant 1; only its existence, positivity and dyadic invariance are used, and these follow from **G** together with the elementary lower bound of Lemma 5.2.